

JEE MAIN REVISION MASTER: WORK POWER & ENERGY

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 153 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Work-Energy Theorem & Conservative Forces	127 PYQs	Work & Energy
Very High	Power & Collisions (Elastic & Inelastic)	35 PYQs	Collisions & Power

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Work-Energy Theorem & Conservative Forces	$W_{\text{net}} = \Delta K$, $W_{\text{ext}} = \Delta K + \Delta U$, $F = -\frac{dU}{dx}$, $U_{\text{spring}} = \frac{1}{2}kx^2$
Power & Collisions (Elastic & Inelastic)	$P = \vec{F} \cdot \vec{v} = \frac{dW}{dt}$, $e = \frac{v_2 - v_1}{u_1 - u_2}$, $\Delta K_{\text{loss}} = \frac{m_1 m_2}{2(m_1 + m_2)}(1 - e^2)(u_1 - u_2)^2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Work-Energy Theorem & Conservative Forces (Very High Yield — 127 PYQs)

Core Principle: Work-energy theorem relating net work to change in kinetic energy, and potential energy gradients for conservative forces.

■■ **Caution / JEE Pitfall:** Work done by static friction on a pure-rolling body is strictly ZERO because the point of contact is instantaneously at rest.

■ **Speed Trick:** Stable equilibrium occurs where $(dU / dx) = 0$ and $(d^2U / dx^2) > 0$ (local potential minimum).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A body of mass 1 kg moves along a straight line with a velocity $v=2x^2$. The work done by the body during displacement from $x=0$ to 5 m..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "A spherical ball of mass 2 kg falls from a height of 10 m and is brought to rest after penetrating 10 cm into sand. The average force exerted..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : An object moves from position r_1 to position r_2 under a conservative force field ϕ ..." [Ans: C]

3.2 Power & Collisions (Elastic & Inelastic) (Very High Yield — 35 PYQs)

Core Principle: Instantaneous power delivery, linear momentum conservation, and coefficient of restitution in 1D and 2D impacts.

■ **Caution / JEE Pitfall:** In perfectly inelastic collisions ($e = 0$), momentum is conserved, but kinetic energy is NOT conserved.

■ **Speed Trick:** For two equal masses undergoing head-on elastic collision ($e = 1$), they completely exchange velocities.

Representative Source PYQs:

- **[2026 Morning Shift] Q#2 (Numerical):** "A body of mass 2 kg begins to move under the influence of time dependent force $\vec{F} = (2t\hat{i} + 6t^2\hat{j})\text{N}$, where $\hat{i}$ and $\hat{j}$..." **[Ans: 200]**
- **[2025 Morning Shift] Q#1 (MCQ):** "An object of mass 1000 g experiences a time dependent force $\vec{F} = (2t\hat{i} + 3t^2\hat{j})\text{N}$. The power generated by the force at time ..." **[Ans: N/A]**
- **[2025 Evening Shift] Q#1 (MCQ):** "A body of mass 4 kg is placed on a plane at a point P having coordinate (3,4)m. Under the action of force $\vec{F} = (2\hat{i} + 3\hat{j})\text{N}$..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

A body of mass 1 kg moves along a straight line with a velocity $v = 2x^2$. The work done by the body during displacement from $x=0$ to 5 m is ____ J.

- (A) 0
- (B) 250
- (C) 1250 **[Correct]**
- (D) 1000

Correct Answer: C

Solution Analysis: Work done (W) is defined as the integral of force (F) over a displacement (dx): $W = \int_{x_i}^{x_f} F dx$. From Newton's Second Law, $F = ma$, and acceleration $a = (dv / dt)$. Using the chain rule, $a = (dv / dx) \cdot (dx / dt) = v(dv / dx)$. $W = \int_{x_i}^{x_f} m(v(dv / dx)) dx \Rightarrow W = \int_{v_i}^{v_f} m v dv = m \int_{v_i}^{v_f} v dv = m [\frac{v^2}{2}]_{v_i}^{v_f} \Rightarrow W = (1/2)mv_f^2 - (1/2)mv_i^2 \Rightarrow W = \Delta \text{K.E.}$ The mass of the body is $m=1\text{kg}$. The velocity of body is given as function of position, $v=2x^2$. Init

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Morning Shift (Q#2)

A body of mass 2 kg begins to move under the influence of time dependent force $\vec{F} = (2t\hat{i} + 6t^2\hat{j})\text{N}$, where $\hat{i}$ and $\hat{j}$ are unit vectors along x and y -axis respectively. The power produced by the force at $t=2\text{s}$ is ____ W.

Correct Answer: 200

Solution Analysis: Power (P) is defined as the rate at which work is done. It can be calculated as the dot product of the force vector ($\vec{F}$) and the velocity vector ($\vec{V}$). According to Newton's Second Law ($\vec{F} = m\vec{a}$), the acceleration is: $\vec{a} = \frac{\vec{F}}{m} = \frac{2t\hat{i} + 6t^2\hat{j}}{2} = (t\hat{i} + 3t^2\hat{j})\text{m/s}^2$. Velocity is the integral of acceleration with respect to time. Since the body begins to move, we assume the initial velocity is zero at $t=0$: $\vec{v} = \int \vec{a} dt = \int_0^t (t\hat{i} + 3t^2\hat{j}) dt = (\frac{t^2}{2}\hat{i} + \frac{3t^3}{3}\hat{j}) = (\frac{t^2}{2}\hat{i} + t^3\hat{j})$